

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Zoology

**Paper: ZOB-301 — Comparative Physiology, Endocrinology and
Developmental Biology
(Old Course 2009-10 Syllabus)**

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 of Section-A, Question No. 1 of Section-B, and Question No. 1 of Section-C which are compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Comparative Physiology) (Marks: 26)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) Define the following: [2 × 1 = 2]
- (i) Ammonotelism
 - (ii) Resting membrane potential.
- (b) State whether the following statements are True or False with proper reasons: [3 × 2 = 6]
- (i) Counter current flow mechanism is found only in fishes.
 - (ii) Shivering is associated with protection against hot environment.
 - (iii) Na⁺-K⁺ pump provides an additional contribution to the resting potential.

Question 2. Answer the following parts: [9 + 9 = 18]

- (a) Describe the functions of various types of leukocytes found in the blood.
- (b) Describe the chemical digestion of proteins in animals.

Question 3. Answer the following parts: [6 + 6 + 6 = 18]

- (a) Discuss the mechanism of branchial respiration in fishes.
- (b) Give an account of various kinds of proteins involved in muscle contraction.
- (c) Describe ornithine cycle of urea formation.

SECTION – B
(Endocrinology) (Marks: 26)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False. Justify your answers: $[4 \times 1 = 4]$

- (i) Zona glomerulosa cells of adrenal gland secrete glucocorticoids.
- (ii) Testosterone is produced by Sertoli cells of the testis.
- (iii) Glucagon is a hyperglycemic hormone.
- (iv) Calcitonin is produced by parafollicular cells of the parathyroid gland.

(b) Define the following: $[4 \times 1 = 4]$

- (i) Endocrine mode of hormone delivery
- (ii) Zona fasciculata
- (iii) Graafian follicle
- (iv) Neurohypophysis.

Question 2. Draw a well labelled diagram of Graafian follicle. Write important biological actions of estrogen. $[4 + 5 = 9]$

Question 3. Describe the structure of the parathyroid gland. Comment upon the role of parathormone in Ca^{2+} homeostasis. $[4 + 5 = 9]$

Question 4. Describe the structure and different parts of pituitary gland and name the hormones secreted by them. $[9]$

SECTION – C
(Developmental Biology) (Marks: 18)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Define the following terms:

[4 × 0.5 = 2]

(i) Acrosome reaction

(ii) Invagination

(iii) Amnion

(iv) Differentiation.

(b) State whether the following statements are True or False. Justify your answer in 2–3 sentences:

[2 × 2 = 4]

(i) Nieuwkoop centre is also called as primary organiser of amphibian.

(ii) The function of the Hensen's node is equivalent to the dorsal lip of the amphibian blastopore.

Question 2. Answer the following parts:

[6 + 6 = 12]

(a) Define the role of organiser in amphibian embryo development.

(b) Define the process of regeneration and describe the limb regeneration in amphibians.

Question 3. Write short notes on any three of the following:

[3 × 4 = 12]

(a) External fertilization

(b) Fate map of chick

(c) Fast block to polyspermy

(d) Primitive streak.

[Click Here](#)

[Click Here](#)

[Click Here](#)